

Figure 1: Autolt recorder window

applications. The official website of AutoIt defines it as a freeware BASIC-like scripting language designed for automating the Windows GUI and general scripting. It basically uses a combination of different keystrokes, mouse movements and window/control manipulation to automate different tasks. AutoIt runs on all versions of Windows. Since its initial release around 17 years ago by Jonathan Bennett and the AutoIt team in January 1999, it has become a very powerful language that supports various complex expressions, user functions, loops and other such features.

The syntax of the scripting language used in AutoIt is similar to that found in the BASIC family of languages. It actually uses C# and VB as its scripting language. AutoIt is a third-generation

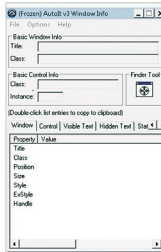


Figure 2: AU3 Info tool

the element's name and other such smart identifiers. All such identifiers' values are stored in a repository in the form of objects, and they are used during playback to identify various elements and perform actions on them.

The AU3 Info object identifier helps retrieve these values for a specific element of an application. It also has several inbuilt function libraries containing predefined functions that must be imported so that different functions can be used in the script. It has a SciTE editor, which

programming language with a classical data model and various data types which can store several types of data. It can be used to automate any specific task by simply writing a few lines of script or even record that process using the AutoIt Recorder.

Once the recording is done, the script is automatically generated and we can use the same to further play back the recorded process. It identifies several elements of the application by using their various properties like the control ID, coordinates,

Table 1: A few open source automation testing tools<sup>1</sup>

| Tool name | Type of applications tested           | Supported scripting language              | Other features                                                                                                                                                                                                                                                                                                                                          |
|-----------|---------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Selenium  | Web applications                      | Java, C#, Ruby, Groovy, PERL, Python, PHP | <ul style="list-style-type: none"> <li>Provides record/playback tool to automate processes</li> <li>Can be deployed on Windows, Linux and Macintosh platforms</li> <li>Identifies components of the application using various element properties</li> <li>Consists of the Selenium IDE, client API, Web driver, grid and remote control</li> </ul>      |
| Autolt    | Windows and desktop applications      | VB and C#                                 | <ul style="list-style-type: none"> <li>Provides record/playback option to automate processes</li> <li>Consists of various inbuilt library functions to perform specific tasks</li> <li>Uses control ID, position coordinates and other such properties to identify elements</li> <li>Keyboard shortcuts can be easily implemented using this</li> </ul> |
| Sikuli    | Windows, desktop and Web applications | Python, Java and Ruby                     | <ul style="list-style-type: none"> <li>Uses image recognition to recognise and control GUI elements</li> <li>Supports scripting with SikuliXIDE</li> </ul>                                                                                                                                                                                              |
| Sahi      | Web applications                      | JavaScript                                | <ul style="list-style-type: none"> <li>Has a record and playback feature to automate various processes</li> <li>Uses Smart Accessor Identification to identify elements with even dynamic IDs</li> <li>Consists of rich inbuilt reports and logs</li> <li>Fast parallel playback feature, which can run several scripts at a time</li> </ul>            |
| SoapUI    | Web applications                      | Groovy script and Java                    | <ul style="list-style-type: none"> <li>Useful for functional Web services, security and load testing</li> <li>Supports data driven testing</li> <li>Supports multiple environments</li> <li>Generates quick reports</li> <li>Provides drag-and-drop option for creating test suites, test steps, etc</li> </ul>                                         |